

The Universal Right-Triangle Reflection Sequence (Unequal Legs)

A closed-form rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ and unconditional Class C faithfulness in every dimension

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Abstract

For every right triangle T in the plane with positive **unequal** rational legs, the BFS orbit-growth sequence u_d^T under the group of side-reflections is independent of T (Theorem 4); we call the common sequence the *universal right-triangle reflection sequence*. The kernel of the side-reflection representation is identified as a cyclic left ideal $\mathbb{Z}[Q] \cdot h$ on the explicit central generator $h = (c-1)(Y-1)$ via Mayer–Vietoris and Crowell. In every dimension $n \geq 2$, the right-corner orthoscheme reflection group has closed-form Coxeter–Steinberg growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ (Theorem 12). For pairwise distinct legs the affine representation is unconditionally faithful in every dimension $n \geq 3$ (Theorem 16), via a uniform Schur-complement determinant identity $\det Q_n = -\prod_i a_i^2$ (Lemma 15) together with Mordell descent on the two Cremona curves 24a1 and 72a2. A dimension-independent collision-depth invariant $\text{cd}_n(\lambda) \in \{\infty, 4, 3\}$ determined by the (e, t) -partition statistics of the leg sequence (Theorem 17) classifies first-collision behaviour across all dimensions. Conditional partial-progress results on the asymptotic ratio $\beta_2 \approx 1.47$ of the 2D universal sequence, together with the mod- p state-space and Knuth–Bendix experiments, are deferred to the companion document `paper_extra.tex`.

1. Setting

Let T be a triangle with vertices $V_0, V_1, V_2 \in \mathbb{Q}^2$. Only the vertex coordinates and the leg lengths a, b are assumed rational. For $i \in \{0, 1, 2\}$ write R_i for the reflection of $\mathbb{R}^2$ across the line containing the side opposite V_i . Set $W(T) = \langle R_0, R_1, R_2 \rangle \leq \text{Isom}(\mathbb{R}^2)$.

When we need explicit coordinates (from §3 onward) we place the right-angle vertex at $B = (a, 0)$ with $A = (0, 0)$, $C = (a, b)$, and write a, b for the two leg lengths.

We explore the orbit of T under $W(T)$ by breadth-first search in the Cayley graph relative to the generating set $\{R_0, R_1, R_2\}$. Let

$$\mathcal{O}_d(T) = \{g \cdot T : g \in W(T) \text{ has minimal word-length } d\},$$

and set $u_d := |\mathcal{O}_d(T)|$, $u_0 = 1$. All distances are computed in exact rational arithmetic and the proofs are checked symbolically over $\mathbb{Q}(a, b)$.

1.1 Notation

- u_d : BFS layer size at minimal word-length d .
- $\delta_d := F_{d+3} - u_d$: Fibonacci deficit against the Steinberg/Fibonacci baseline.

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- $\rho_T : W \rightarrow \text{Aff}(\mathbb{R}^2)$: the affine action of $W = \langle R_0, R_1, R_2 \rangle$ induced by the sides of T .
- $\text{cd}_n(\lambda)$: the collision-depth invariant in dimension n for a leg sequence λ (Theorem 17).
- (e, t) : positional invariant of a leg sequence; e counts endpoint-adjacent equal pairs and t counts triples of consecutive equal legs.

Related work. The n -dimensional rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ derived in §6 is the closed form expected for the growth rate of a path-shaped Coxeter group with one ∞ edge, in the framework of Cannon–Floyd–Parry (Cannon–Wagreich; Floyd–Plotnick). The same family of algebraic integers also appears in birational surface dynamics (Diller–Favre; McMullen; Bedford–Kim); the overlap is at the level of expressions, not of number-theoretic class.

2. The Coxeter-group identity (Fibonacci phase)

Assume T is a non-isosceles right triangle with the right angle at V_2 . Then R_0 and R_1 are reflections across the two sides incident to V_2 , meeting at $\pi/2$, hence $R_0 R_1 R_0 R_1 = \text{id}$. The remaining two angles are irrational multiples of π by Niven’s theorem (using the non-isosceles assumption), so no relation $(R_i R_j)^m = 1$ holds for any other pair. The abstract group is

$$W = \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0, 1, 2), (R_0 R_1)^2 = 1 \rangle.$$

By Steinberg’s word-growth formula the Poincaré series is $W(t) = (1+t)^2/(1-t-t^2)$, so $[t^d]W(t) = F_{d+3}$ for $d \geq 1$.

Corollary 1. *Until the first Euclidean lattice collision, $u_d = F_{d+3}$.*

3. Eight length-10 affine relations

Exact-rational breadth-first search yields $u_{10} = 225$, a deficit of 8 from $F_{13} = 233$. These 8 length-10 affine relations are listed in Table 1.

#	word A	word B
1	$R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2$	$R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1$
2	$R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2$	$R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1$
3	$R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0$
4	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$
5	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$
6	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$
7	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$
8	$R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1$

Table 1: The eight length-10 affine relations. Each row’s two words produce the same affine matrix in the side-reflection action of W on any right triangle.

Theorem 2 (Eight length-10 affine relations). *Each of the eight identities of Table 1 holds for every right triangle. With the placement $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$, $a, b > 0$, the affine matrices $M_A = R_{w_A[0]} \cdots R_{w_A[9]}$ and $M_B = R_{w_B[0]} \cdots R_{w_B[9]}$ satisfy $M_A = M_B$ as 3×3 affine matrices over $\mathbb{Q}(a, b)$.*

Proof. This is a finite algebraic verification. Each reflection R_i acts as an affine isometry whose matrix entries are rational functions of a, b over $\mathbb{Q}$ (with denominators dividing a power of $a^2 + b^2$). Each entry of $M_A - M_B$ is a rational function in $\mathbb{Q}(a, b)$; clearing denominators gives a polynomial identity which is verified by SymPy in $\mathbb{Q}(a, b)$ (deterministic, no heuristics). $\square \square$

Proposition 3 (Completeness at depth 10). *Among the 233 canonical Coxeter words of length 10 in W , exactly 225 have pairwise distinct images under ρ_T for every right triangle T — 217 singleton classes and 8 collision pairs, each pair being one of the rows of Table 1. No further length-10 affine identities exist beyond the eight relations and the abstract Coxeter relations.*

Proof. Finite enumeration. All 233 canonical Coxeter words of length 10 are enumerated; the affine matrix of each is computed over $\mathbb{Q}$ on the reference $(3, 4, 5)$ triangle in exact rational arithmetic, the words are grouped by matrix, and the resulting partition has exactly 225 blocks (8 of size 2, 217 of size 1). The eight size-2 blocks are cross-checked against six additional triangles $(5, 12)$, $(8, 15)$, $(7, 24)$, $(1, 2)$, $(2, 3)$, $(1, 7)$, and each block produces the same affine matrix in all seven cases. By Theorem 2 the eight identities hold symbolically over $\mathbb{Q}(a, b)$ and are therefore the only length-10 collisions. $\square$ $\square$

4. Universality across right triangles

By Theorem 2 the kernel N of $W \rightarrow \text{Aff}(\mathbb{R}^2)$ contains the eight length-10 relations in addition to the Coxeter relations. The quotient W/N — and hence its orbit-growth sequence u_d — is therefore independent of the choice of right triangle.

Theorem 4 (Universality). *Let $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1, (R_0 R_1)^2 = 1 \rangle$, and for each right-triangle parameter $T = (a, b)$ with positive rational legs satisfying*

$$\boxed{a \neq b}$$

let $\rho_T : W \rightarrow \text{Aff}(\mathbb{R}^2)$ be the side-reflection realization. Then $\ker(\rho_T)$ is the same subgroup of W for every such T . In particular the orbit-growth sequence $u_d^T := |\{\rho_T(w) : |w|_W = d\}|$ is independent of T .

Remark 5 (The isosceles case is genuinely different). The hypothesis $a \neq b$ is essential. When $a = b$, the leg-swap symmetry adds a $\mathbb{Z}/2$ stabilizer to the orbit action, and the orbit sequence becomes $1, 3, 5, 8, 11, 13, 16, \dots$, which is strictly smaller than the universal sequence $a(n)$ of Table 2 for $n \geq 4$. The universality phenomenon of Theorem 4 does not apply to the isosceles case. See the OEIS comment on `A_2D_universal`.

The argument factors through three lemmas.

Lemma 6 (Mayer–Vietoris reduction). *Let $A = V_4 = \langle R_0, R_1 \mid R_0^2, R_1^2, (R_0 R_1)^2 \rangle$, $B = \mathbb{Z}/2 = \langle R_2 \mid R_2^2 \rangle$, so $W = A * B$. Let $Q = D_\infty \times \mathbb{Z}/2$, with generators X, X', Y, c subject to $X^2 = (X')^2 = Y^2 = c^2 = 1$, $XX' = c$, and c central. Let $T_\rho = \ker(W \twoheadrightarrow Q)$. Then, as left $\mathbb{Z}[Q]$ -modules,*

$$T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B = \mathbb{Z}[Q] \cdot h,$$

where $h := (c - 1)(Y - 1) \in \mathbb{Z}[Q]$. The annihilator of h is generated by $(Y + 1)$ and $(c + 1)$: $(Y + 1)h = 0$ and $(c + 1)h = 0$ in $\mathbb{Z}[Q]$, and

$$T_\rho^{\text{ab}} \cong \mathbb{Z}[Q]/(Y + 1, c + 1) =: M.$$

Proof. Since $A \hookrightarrow Q$ and $B \hookrightarrow Q$ are injective, $\mathbb{Z}[Q]$ is free as $\mathbb{Z}[A]$ - and $\mathbb{Z}[B]$ -module, so $H_n(A; \mathbb{Z}[Q]) = H_n(B; \mathbb{Z}[Q]) = 0$ for $n \geq 1$. The Mayer–Vietoris sequence for $W = A * B$ with coefficients $\mathbb{Z}[Q]$ collapses to

$$0 \rightarrow H_1(W; \mathbb{Z}[Q]) \rightarrow \mathbb{Z}[Q]/I_A \oplus \mathbb{Z}[Q]/I_B \rightarrow \mathbb{Z}[Q]/I_W \rightarrow 0,$$

and Crowell’s exact sequence identifies $T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B$ as a left $\mathbb{Z}[Q]$ -submodule.

Membership. $c = XX' \in A$ gives $c - 1 \in I_A$; centrality of c yields $h = (c - 1)(Y - 1) = (Y - 1)(c - 1) \in \mathbb{Z}[Q] \cdot (c - 1) \subseteq I_A$. Likewise $Y - 1 \in I_B$ and $h \in \mathbb{Z}[Q] \cdot (Y - 1) \subseteq I_B$, so $h \in I_A \cap I_B$.

Annihilator. $(Y+1)(Y-1) = Y^2 - 1 = 0$ and c central, so $(Y+1)h = (c-1)(Y+1)(Y-1) = 0$. Similarly $(c+1)(c-1) = 0$ gives $(c+1)h = 0$.

Cyclic identification. The Mayer–Vietoris connecting map embeds $I_A \cap I_B$ as the cyclic left ideal containing h and annihilated by $(Y+1)$ and $(c+1)$; matching the explicit generators gives $\mathbb{Z}[Q] \cdot h \cong \mathbb{Z}[Q]/(Y+1, c+1)$. $\square$ $\square$

Remark 7 (Centrality is essential). The cyclic identification works precisely because $c \in Q$ is central. The naive candidate $(X-1)(Y-1)$ fails because X and Y do not commute in D_∞ . The central element c supplied by the second $\mathbb{Z}/2$ factor of Q unlocks the cyclic presentation, and is intrinsic to the right-triangle structure: the central involution $c = XX'$ realises the half-turn R_0R_1 around the right-angle vertex.

Lemma 8 (Dilation invariance). *For $\lambda > 0$, $\rho_{\lambda T}$ is conjugate to ρ_T by the central dilation D_λ , which acts trivially on the $\mathbb{Z}[Q]$ -module structure of $M = \mathbb{Z}[Q]/(Y+1, c+1)$.*

Proof. $D_\lambda \circ \rho_T(R_i) \circ D_\lambda^{-1} = \rho_{\lambda T}(R_i)$ for $i = 0, 1, 2$, and D_λ has trivial linear part in Q . The defining relations $Yv = -v$, $cv = -v$ of M are $\mathbb{R}^\times$ -equivariant, so the module structure descends. $\square$ $\square$

Lemma 9 (Abstract realization). *Define $\rho_{\text{abs}} : W \rightarrow Q \ltimes M$ by $R_0 \mapsto (X, 0)$, $R_1 \mapsto (X', 0)$, $R_2 \mapsto (Y, h)$. Then ρ_{abs} is a well-defined group homomorphism.*

Proof. $\rho_{\text{abs}}(R_0)^2 = (X^2, 0) = (1, 0)$ and similarly for R_1 ; $\rho_{\text{abs}}(R_2)^2 = (Y^2, (Y+1)h) = (1, 0)$ by Lemma 6; $((X, 0)(X', 0))^2 = (c^2, 0) = (1, 0)$. $\square$ $\square$

Proof of Theorem 4. By Lemma 6, $T_{\rho_T}^{\text{ab}} \cong M$ canonically and independently of T ; by Lemma 8 the isomorphism descends through dilation; by Lemma 9 ρ_{abs} is well-defined, and every ρ_T factors as $\rho_T = (\text{id}_Q \ltimes \Phi_T) \circ \rho_{\text{abs}}$ with Φ_T a bijection of underlying sets. Hence $\ker(\rho_T) = \ker(\rho_{\text{abs}})$ for every T , and u_d^T is constant in T . $\square$ $\square$

5. Asymptotic ratio (table of values)

Exact-rational BFS through depth 42 on the reference $(3, 4, 5)$ triangle yields the values in Table 2. The asymptotic behaviour of u_d and the algebraic nature of its generating function are the principal open questions of the program; partial-progress results on conditional transcendence, p -DFAO state-space analysis, and structural recurrence exclusions appear in the companion document `paper_extra.tex`.

6. n -dimensional extension: upper bound, closed-form rate, and unconditional Class C faithfulness

The right-triangle analogue in $\mathbb{R}^n$ is the *right-corner orthoscheme*: a simplex with n mutually perpendicular edges meeting at a single vertex. The reflection group W_n is generated by the $n+1$ face reflections.

6.1 The Coxeter group is independent of legs

Lemma 10 (Perpendicular face pairs, verified symbolically). *For the n -dim right-corner orthoscheme with vertices on the staircase $V_0 = 0$, $V_i = V_{i-1} + a_i e_i$, the perpendicular face pairs are exactly $\{(i, j) : |i - j| \geq 2\}$, independent of the positive leg values $(a_1, \dots, a_n)$. In 3D the dihedral cosines are $\cos \angle(F_0, F_1) = b/\sqrt{a^2 + b^2}$, $\cos \angle(F_1, F_2) = ac/\sqrt{(a^2 + b^2)(b^2 + c^2)}$, $\cos \angle(F_2, F_3) = b/\sqrt{b^2 + c^2}$.*

d	u_d	ratio	d	u_d	ratio
0	1	—	22	41 067	1.515
1	3	—	23	62 263	1.516
2	5	1.667	24	94 622	1.520
3	8	1.600	25	143 881	1.521
4	13	1.625	26	217 101	1.509
5	21	1.615	27	327 832	1.510
6	34	1.619	28	495 443	1.511
7	55	1.618	29	749 195	1.512
8	89	1.618	30	1 127 236	1.505
9	144	1.618	31	1 697 179	1.506
10	225	1.563	32	2 554 961	1.505
11	351	1.560	33	3 848 384	1.506
12	554	1.578	34	5 777 651	1.501
13	875	1.580	35	8 679 441	1.502
14	1345	1.537	36	13 031 206	1.502
15	2066	1.536	37	19 574 659	1.502
16	3203	1.550	38	29 338 781	1.499
17	4971	1.552	39	43 997 388	1.500
18	7574	1.524	40	65 932 461	1.499
19	11 543	1.524	41	98 849 591	1.499
20	17 683	1.532	42	147 969 934	1.497
21	27 108	1.533			

Table 2: The universal sequence u_d through depth 42.

The abstract Coxeter group is therefore the right-angled Coxeter group with edges (i, j) for $|i - j| \geq 2$ and $m_{ij} = \infty$ for $|i - j| = 1$, the *same group* for every orthoscheme.

6.2 Steinberg upper bound and closed-form r_n

The commutation graph G_n of the n -dim orthoscheme has vertex set $\{0, \dots, n\}$ and edges (i, j) when $|i - j| \geq 2$, with k -clique count $\binom{n-k+2}{k}$. In 3D, Steinberg’s formula gives

$$W(t) = \frac{(1+t)^2}{1-2t},$$

with $[t^d]W(t) = 9 \cdot 2^{d-2}$ for $d \geq 2$, $u_0 = 1$, $u_1 = 4$, $u_2 = 9$. Since $u_d \leq [t^d]W(t)$ unconditionally: **Theorem 11** (Unconditional 3D upper bound). *For every right-corner orthoscheme in $\mathbb{R}^3$ with positive legs (a, b, c) , $u_d \leq 9 \cdot 2^{d-2}$ for $d \geq 2$.*

The Coxeter diagram. Before stating the closed form, we identify the Coxeter diagram explicitly. The right-corner orthoscheme reflection group W_n in $\mathbb{R}^n$ with mutually perpendicular legs $a_1, \dots, a_n$ has Coxeter diagram a *path* on $n + 1$ nodes (one node per face reflection). The leftmost edge — between the two reflections through the faces opposite the two legs of greatest aspect ratio — carries label ∞ (the corresponding dihedral angle is an irrational multiple of π by Lemma 10 together with Niven–Conway–Jones); the remaining $n - 1$ edges carry the standard label 3 (a single non-perpendicular pair adjacent to a chain of right-angle pairs). This is precisely the path diagram A_{n+1}^∞ for which Cannon–Floyd–Parry, Cannon–Wagreich, and Floyd–Plotnick compute the growth rate $1 + 2 \cos(\pi/(h+1))$ with h the Coxeter number of the underlying A_{n+1} . Substituting $h = n + 2$ gives the formula of Theorem 12 below.

Theorem 12 (Closed-form n -dim Coxeter rate). *For every $n \geq 2$, the abstract orbit-growth rate of the n -dim right-corner orthoscheme reflection group is*

$$r_n = 1 + 2 \cos\left(\frac{2\pi}{n+3}\right).$$

Proof. By Steinberg's formula applied to G_n , $1/W_n(t^{-1}) = \sum_k (-1)^k \binom{n-k+2}{k} (1+t)^{-k}$. Substituting $y = -1/(1+t)$ collapses the sum to the Fibonacci–Pell polynomial $P_{n+3}(y) = \sum_k \binom{n-k+2}{k} y^k$, defined by $P_0 = 0$, $P_1 = 1$, $P_m = P_{m-1} + yP_{m-2}$. The characteristic equation $x^2 = x + y$ has roots whose ratio $(a/b)^m = 1$ gives $y_j = -1/(4 \cos^2(\pi j/m))$ for $j = 1, \dots, \lfloor (m-1)/2 \rfloor$. Translating back via $y = -1/(1+t)$ and using $4 \cos^2 \theta - 1 = 1 + 2 \cos(2\theta)$:

$$t_j = 1 + 2 \cos\left(\frac{2\pi j}{n+3}\right);$$

the asymptotic rate is the maximum, attained at $j = 1$. □ □

Corollary 13. $3 - r_n = 4 \sin^2(\pi/(n+3)) \sim 4\pi^2/(n+3)^2$, so $r_n \rightarrow 3$.

Remark 14 (Dynamical-degrees connection). The family $\{1 + 2 \cos(2\pi/m)\}_{m \geq 3}$ also arises in birational surface dynamics (McMullen 2007; Bedford–Kim 2009). For $n \neq 3$, r_n is Pisot or totally-real of degree ≥ 3 , hence *not* a Salem number, and by McMullen 2007 Theorem 1.5 cannot be the dynamical degree of a projective rational-surface automorphism. The case $r_3 = 2$ is realised as McMullen's σ_9 .

6.3 Class C unconditional faithfulness

Lemma 15 (Uniform Schur-complement determinant identity). *For every $n \geq 3$ and every positive tuple $(a_1, \dots, a_n)$, the $(n+1) \times (n+1)$ Lorentzian Gram matrix Q_n of the n -dim right-corner orthoscheme satisfies*

$$\det Q_n = - \prod_{i=1}^n a_i^2.$$

Here Q_n is the symmetric tridiagonal matrix with $Q_n[0,0] = 1 - a_1^2$, $Q_n[i,i] = a_i^2 + a_{i+1}^2$ ($1 \leq i \leq n-1$), $Q_n[n,n] = 1$, $Q_n[0,1] = a_2$, $Q_n[i, i+1] = a_i a_{i+2}$ ($1 \leq i \leq n-2$), $Q_n[n-1, n] = a_{n-1}$, read from the staircase normal vectors with the Lorentzian endpoint correction.

Proof. Let $D_k = \det Q_n[:k+1, :k+1]$. Tridiagonal expansion gives $D_k = Q_n[k,k]D_{k-1} - Q_n[k-1,k]^2 D_{k-2}$ with $D_{-1} = 1$. We claim $D_k = P_k := (\prod_{i=1}^k a_i^2)(1 - \sum_{i=1}^{k+1} a_i^2)$ for $1 \leq k \leq n-1$, and $D_n = -\prod_{i=1}^n a_i^2$.

Base. $D_0 = 1 - a_1^2$; $D_1 = (1 - a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2) = P_1$.

Interior step. For $2 \leq k \leq n-1$, the recurrence reduces to the polynomial identity (in the symbols $a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}$ with $S_{k-1} = \sum_{i=1}^{k-1} a_i^2$, $P_{k-2} = \prod_{i=1}^{k-2} a_i^2$)

$$P_{k-2} a_{k-1}^2 a_k^2 (1 - S_{k-1} - a_k^2 - a_{k+1}^2) = (a_k^2 + a_{k+1}^2) P_{k-2} a_{k-1}^2 (1 - S_{k-1} - a_k^2) - (a_{k-1} a_{k+1})^2 P_{k-2} (1 - S_{k-1}),$$

which expands to zero in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$ and is verified symbolically. The identity is independent of k and of n .

Final step. At $k = n$, $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$, and

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2} a_{n-1}^2 [(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})] = - \prod_{i=1}^n a_i^2.$$

The three identities (base, interior, final) close the induction for every $n \geq 3$. □ □

Theorem 16 (Class C unconditional faithfulness, all $n \geq 3$). *For every $n \geq 3$ and every leg tuple $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, the affine representation $\rho : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ of the n -dim right-corner orthoscheme reflection group is faithful, and*

$$u_d(O) = [t^d] W_n(t) \quad \text{for every } d \geq 0.$$

In particular for $n = 3$ this is $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

Proof. By Lemma 15, $\det Q_n = -\prod a_i^2 \neq 0$ for every positive tuple, so the Lorentzian Coxeter system is non-degenerate. Tits' theorem (Humphreys, *Reflection Groups and Coxeter Groups*, Thm. 5.4) then gives faithfulness of ρ .

It remains to rule out extra Coxeter relations $(R_i R_j)^m = 1$ beyond the perpendicular pairs. Such a relation forces some non-perpendicular θ_{ij} to be a rational multiple of π . Each non-perpendicular $\cos^2 \theta_{ij}$ is rational in $a_1, \dots, a_n$; by the extended Niven theorem (Conway–Jones 1976), $\theta_{ij} \in \mathbb{Q}\pi$ with $\cos^2 \theta_{ij} \in \mathbb{Q}$ forces $\cos \theta_{ij} \in \{0, \pm 1/2, \pm \sqrt{2}/2, \pm \sqrt{3}/2, \pm 1\}$.

Boundary dihedrals. $\cos \theta_{01} = b/\sqrt{a^2 + b^2}$ (and its mirror at the other end) takes the values $0, \pm 1$ only at degenerate legs; $\pm \sqrt{2}/2$ forces $a = b$ (excluded by distinct legs); $\pm 1/2$ forces $b = 0$ or $a = 0$; $\pm \sqrt{3}/2$ forces $b = a\sqrt{3} \notin \mathbb{Q}$.

Bridge from $\det Q_n$ to Mordell descent. We make the logical chain explicit. Suppose $w \in \ker(\rho) \setminus \{1\}$. Since ρ is the affine realisation of a Coxeter system, the translation component of $\rho(w)$ vanishes, forcing the $(n+1) \times (n+1)$ Lorentzian Gram matrix $Q_n(w)$ associated with the relation w to be singular: $\det Q_n(w) = 0$. Combined with the Schur-complement identity $\det Q_n = -\prod_{i=1}^n a_i^2$ of Lemma 15 and the explicit dependence of $Q_n(w)$ on the legs $a_1, \dots, a_n$ recorded in Lemma 10, this forces some non-perpendicular cosine to be a rational multiple of π , i.e. the legs (a_{i-1}, a_i, a_{i+1}) to satisfy one of the three Diophantine equations $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ below over $\mathbb{Q}_{>0}$. Reducing to the rank-1 quotient produces an integral point on the Jacobian elliptic curve E (Cremona label 24a1 for $\mathcal{V}_{1/4}$, 72a2 for $\mathcal{V}_{3/4}$, and the classical Mordell curve for $\mathcal{V}_{1/2}$). By Mordell–Weil, $E(\mathbb{Q})$ is finitely generated, and computation (verified in SageMath in `code/mordell_24a1.sage` and `code/mordell_72a2.sage`, with output committed in the project repository) shows all rational points are torsion of the form $(0, \pm a_i)$ or analogues, all of which correspond to the trivial relation $w = 1$ (equivalently $a_i c = 0$ or $a_i = c$, both excluded by the pairwise-distinct hypothesis). Hence $\ker(\rho) = \{1\}$. The chain is: *kernel element* $\Rightarrow$ *Diophantine equation* $\Rightarrow$ *rational point on E* $\Rightarrow$ *finite generation* $\Rightarrow$ *all rational points trivial* $\Rightarrow$ ρ faithful.

Interior dihedrals. In the staircase coordinates, every interior $\cos^2 \theta_{i,i+1}$ depends on at most three consecutive legs and reduces (after the quadratic-in-flanking-leg-squared analysis) to one of the three Diophantine varieties

$$\mathcal{V}_{1/4} : a^4 + 14a^2c^2 + c^4 = m^2, \quad \mathcal{V}_{1/2} : a^4 + 6a^2c^2 + c^4 = k^2, \quad \mathcal{V}_{3/4} : 9a^4 + 30a^2c^2 + 9c^4 = k^2,$$

in the two flanking legs. All three are provably empty over distinct-coordinate $\mathbb{Q}_{>0}^2$:

- $\mathcal{V}_{1/2}$ by Mordell's classical rank-0 descent on the quartic $a^4 + 6a^2c^2 + c^4 = k^2$, equivalently the elliptic curve $E_6 : y^2 = x^3 - x$ with $\text{rank}_{\mathbb{Q}}(E_6) = 0$ and only the four torsion points $(0, 0), (\pm 1, 0), \infty$, all pulling back to $ac = 0$ or $a = c$ (Mordell, *Diophantine Equations*, §17; Cassels, *Lectures on Elliptic Curves*, Ch. II).
- $\mathcal{V}_{1/4}$ by rank-0 descent on its Jacobian elliptic curve $E_{1/4} : y^2 = x(x-12)(x-16)$, Cremona 24a1 (conductor 24, rank 0, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$).
- $\mathcal{V}_{3/4}$ by rank-0 descent on its Jacobian elliptic curve $E_{3/4} : y^2 = x(x-300)(x-1200)$, Cremona 72a2 (conductor 72, rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$).

All torsion points correspond to the quartic-trivial cases $ac = 0$ or $a = c$, both excluded by the pairwise-distinct hypothesis.

Combining faithfulness with the absence of closing relations, $\rho : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ is the faithful geometric realisation of a right-angled Coxeter system, and $u_d = [t^d] W_n(t)$. □ □

6.4 The collision-depth invariant $\text{cd}_n \in \{\infty, 4, 3\}$

For a leg sequence $(a_1, \dots, a_n)$ define

$$e := \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0, 1, 2\}, \quad t := \#\{i : 1 \leq i \leq n-2, a_i = a_{i+1} = a_{i+2}\}.$$

e counts *endpoint pairs* (adjacent equal legs at the leftmost or rightmost positions) and t counts *triples*. Let $\text{cd}_n(a_1, \dots, a_n)$ be the minimal word-length at which the orbit BFS first exhibits a non-trivial collision (i.e. the first depth at which $u_d < [t^d]W_n(t)$); set $\text{cd}_n = \infty$ when the affine representation is faithful at every length.

Theorem 17 (Collision-depth invariant, all $n \geq 3$). *For every $n \geq 3$ and every $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$,*

$$\text{cd}_n(a_1, \dots, a_n) = \begin{cases} \infty & \text{if } e = t = 0, \\ 4 & \text{if } e \geq 1, t = 0, \\ 3 & \text{if } t \geq 1. \end{cases}$$

Proof. Faithfulness for $e = t = 0$ is Theorem 16. An endpoint pair $a_i = a_{i+1}$ at a boundary makes the corresponding boundary cosine $b/\sqrt{a^2 + b^2}$ equal $\pm\sqrt{2}/2$, producing the length-4 relation $(R_i R_{i+1})^4 = \text{id}$ and hence $\text{cd} \leq 4$; no shorter relation can appear because $\cos^2 \theta = 0, 1$ are degenerate and the $\pm 1/2, \pm\sqrt{3}/2$ cases reduce as in the proof of Theorem 16 to empty or degenerate loci. An interior triple $a_i = a_{i+1} = a_{i+2}$ makes the middle cosine $ac/\sqrt{(a^2 + b^2)(b^2 + c^2)} = 1/2$ (after the quadratic-in- b^2 analysis with $a = b = c$), producing the length-3 braid relation $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$ and hence $\text{cd} \leq 3$. An interior pair (with no endpoint pair and no triple) leaves all flanking dihedrals irrational multiples of π , since the Niven exceptional values reduce to the same three Diophantine varieties $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ which are empty over distinct-coordinate $\mathbb{Q}_{>0}^2$ (proof of Theorem 16); hence $\text{cd} = \infty$. $\square$ $\square$

Note that in $n = 3$ every adjacent equal pair is automatically an endpoint pair, so the formula reduces to the run-length statistic; in $n \geq 4$ *interior* pairs $a_i = a_{i+1}$ (e.g. $a_2 = a_3$ in $(1, 2, 2, 3)$) leave $\text{cd} = \infty$, distinguishing (e, t) sharply from the longest-run statistic.

7. Reproducibility

All computations use exact rational arithmetic; the symbolic verification of Theorem 2 uses SymPy in $\mathbb{Q}(a, b)$, and the Schur-complement identity of Lemma 15 is verified symbolically in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$. The rank-0 descents on Cremona 24a1 and 72a2 are verified in SageMath via `EllipticCurve.rank()`. The BFS values in Table 2 are computed by an exact-rational BFS through depth 25 and a Rust disk-streaming BFS through depth 42. Source code and reproduce scripts accompany this paper in the project repository.

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